Scanner-Invariant and Institution-Invariant Deep Learning for Reliable Medical Image Diagnosis Across Heterogeneous Clinical Environments

What Invariance Methods Do and Do Not Deliver Under Leave-Site-Out Evaluation, and How to Calibrate Prediction Sets for Unseen Sites

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Abstract

Diagnostic image models often get worse at hospitals that were not part of their training data. We tried to build a scanner- and institution-invariant pipeline from three parts: a class-conditional site adversary, augmentation based on measured per-site image statistics, and conformal prediction sets meant to hold up at an unseen site. We tested it on an open chest-radiograph benchmark (924 images, 10 sources, COVID-19 versus other findings), holding out one site at a time. Four sites guided every design decision. The other four were run once, with settings frozen beforehand. The invariance ideas did not help. Over eight held-out sites and nine training methods, none beat plain empirical risk minimization (ERM) by more than noise: the largest gains were +0.015 AUROC (GroupDRO) and +0.011 (one augmentation-plus-adversary variant), and every 95% interval includes zero. The useful findings concern evaluation and uncertainty. Five simple acquisition statistics predict the diagnosis on unseen sites (AUROC 0.662). Validation on sites seen in training overstates held-out AUROC by 0.118, and standard conformal sets cover only 0.836 of cases at a 0.90 target. A leave-site-out CV+ calibration removed the worst shortfalls at the cost of sets about 26% larger; all calibrations we tried follow one coverage–size curve. Simulated acquisition shift did not reproduce the real under-coverage. The data are small and cover one modality, so we make no claim about clinical COVID-19 detection.

1 Introduction

A diagnostic model trained on pooled data from several hospitals can learn cues that identify the hospital instead of the disease [8, 10, 32]. It then looks good on validation data from the same hospitals and fails quietly at a new one. A common response is to make the model invariant to scanner and institution, through adversarial alignment, feature-statistics matching, harmonization or augmentation [9, 12, 26, 33].

We started this project to build such a recipe. It had three parts. First, a site adversary that works within each class, so that different disease prevalence across hospitals is not penalized [34]. Second, augmentation that re-renders an image as if it came from another *observed* site, using statistics estimated from the training images without labels. Third, conformal prediction sets calibrated with an unseen site in mind. We then ran what we thought would be the deciding experiment: leave-one-site-out (LOSO) testing with a development/confirmation split and configurations frozen before the confirmation run.

The recipe did not work. No method, ours or the baselines, was reliably better than ERM. What survived are results about *why*, and about reporting uncertainty honestly for sites the model has not seen. We report both, since a frozen negative result is useful to anyone choosing among these methods on small multi-site data. Our contributions are modest, and we mark novelty carefully (our literature search was targeted, not systematic):

- a frozen LOSO protocol on open data with a hierarchical paired bootstrap, and a negative result for nine methods, three of them our own variants, plus a controlled label-shift sweep;
- two label-free diagnostics of acquisition shortcuts, one measuring leakage of the label and one measuring how much a model leans on it;
- evidence on calibrating prediction sets for unseen sites: seen-site validation is optimistic, standard conformal sets under-cover, a site-fold CV+ rule avoids the worst cases, and simulated shift does not reproduce the real effect.

We do not claim clinical COVID-19 detection performance; the benchmark is a shortcut-prone stress test.

2 Related Work

Shortcuts in chest radiography. Pneumonia and COVID-19 classifiers depend on site and dataset artifacts [21, 32]. Masking the lungs barely changes accuracy on public COVID-19 collections [18], classifiers can prefer shortcuts to signal [8], and a large review found bias throughout this literature [22, 27]. Our diagnostics follow this line and add a held-out-site measurement.

Invariance and domain generalization. DANN, conditional adversaries, CORAL, GroupDRO, IRM and MixStyle target shift [2, 9, 17, 25, 26, 35]. DomainBed found that, with fair model selection, no method reliably beats ERM [11], and WILDS reports similar difficulty on real shifts [13]. Exact invariance is provably limited when label priors differ across domains [34]; methods that handle joint conditional and label shift assume target data [14, 15]. We study the setting with no target data. Augmentation work (BigAug, stain augmentation, ComBat) is the closest relative of our site-profile augmentation [12, 28, 33].

Conformal prediction under shift. Split conformal prediction needs exchangeability [1, 31]. Weighted and non-exchangeable versions handle known covariate shift, drift, label shift or several sources [4, 16, 20, 29]. For unseen domains, worst-case calibration over source domains or divergence balls exists [36], and per-domain robust calibration is a baseline in set-valued domain generalization [30]; coverage failures of conformal sets under domain shift in medical imaging are documented [19]. Our worst-site mode is a variant of these and not new. CV+ comes from the jackknife+ [3]. We did not find an evaluation of CV+ with sites as folds on real multi-site medical images, but we did not search exhaustively.

3 Methods

Setting. Data come from K source sites that differ in image acquisition $P_k(X \mid Y)$ and prevalence $P_k(Y)$. Models are trained, selected and calibrated on source sites, then tested on one held-out site. The task is binary (COVID-19 vs. other). Write z for the encoder output and s for the site index.

Training methods. All methods share one encoder, optimizer and budget. Baselines: ERM, GroupDRO (sites as groups), DANN (unconditional site adversary, weight 0.5), and generic augmentation (intensity, gamma, noise, blur and bias-field changes applied per sample). The variants we implemented:

- *Class-conditional adversary.* One site discriminator per class behind a gradient-reversal layer, with each sample sent to the discriminator of its label: $\mathcal{L}_{\text{adv}} = \mathbb{E}\,\text{CE}(d_y(\text{GRL}_\lambda(z)), s)$. A discriminator that only sees one class cannot exploit prevalence differences. We pair it with per-class CORAL and site-prior reweighting of the task loss, $w_{s,y} = \pi(y)/\pi_s(y)$.
- *Site-profile augmentation.* We measure five statistics per image (mean, contrast, noise floor, sharpness, low-frequency bias), summarize them per training site without labels, and translate an image from its site toward another observed site by applying only the relative change. It is a measurement-driven proxy, not a scanner simulator.
- *Profile-adversarial invariance (PAI).* A gradient-reversal head tries to regress the measured profile of the input from z.
- *SIDL variants.* SIDL-cc: generic augmentation, class-conditional adversary, per-class CORAL, prior reweighting, and a consistency loss between clean and augmented views. SIDL-phys: site-profile augmentation, unconditional adversary, pooled CORAL, consistency. SIDL-full: site-profile augmentation plus the SIDL-cc components.

Mild settings (augmentation probability 0.3, strength 0.5, adversary weight 0.1 for SIDL and PAI) were chosen on development folds and frozen.

Diagnostics. *Leak:* a logistic regression on the five profile numbers is trained on source sites and scored on the held-out site; AUROC above 0.5 means acquisition statistics alone, without anatomy, predict the label. *Reliance:* within each class, the Spearman correlation between a model's held-out P(COVID-19) and that profile-only score, averaged over classes; working within class means the label cannot explain the correlation.

Conformal calibration. We use the score $1 - p_\theta(y \mid x)$ [23] and sets $\{y : 1 - p_\theta(y \mid x) \le \hat{q}\}$. Three standard variants use the validation split of seen source sites: pooled, site-balanced (equal weight per site) and worst-site (largest per-site quantile among sites with at least 25 images). For the leave-site-out (LSO) variants, we train an inner model for every source site k on the other source sites and score *all* images of site k with it, giving scores R_i that mimic behavior on an unseen site. LSO-split applies the pooled quantile of the R_i to the deployed model.

Source	Images	COVID-19	Other	Prevalence	Role
Actualmed	185	58	127	0.31	development
China	56	51	5	0.91	development
Italy	120	100	20	0.83	development
Spain	56	38	18	0.68	development
Australia	60	6	54	0.10	confirmation
Other	207	122	85	0.59	confirmation
UK	66	49	17	0.74	confirmation
Unknown	52	33	19	0.63	confirmation
Figure1	40	35	5	0.88	training only
Hannover-DE	82	79	3	0.96	training only
Total	924	571	353	0.62	

Table 1: Benchmark composition. Role is the fold role when the source is held out; Figure1 and Hannover-DE have too few negatives and are only used for training.

LSO-worst takes the largest per-site quantile. LSO-CV+ applies the CV+ rule with sites as folds: label y enters the set for test image x iff $\#\{i : R_i < 1 - p_{-k(i)}(y \mid x)\} < (1-\alpha)(n+1)$, where every inner model scores x. CV+ guarantees coverage for exchangeable *points* [3]; here the exchangeable units would be sites, we have only nine, and we have no theorem. Coverage is measured empirically. Besides coverage and mean set size at $\alpha = 0.1$, we report the coverage-calibration error (mean $|\text{coverage} - (1-\alpha)|$ over nine values of α from 0.04 to 0.30), the fraction of (site, α) cells more than five points under target, and set size interpolated at 90% coverage.

4 Benchmark and Protocol

Data. We pooled three public GitHub collections of frontal chest radiographs [5–7]. A site is the institution or country in the metadata. Small sites are merged into "Other" and images with no location form "Unknown", so these two groups mix institutions. The label is COVID-19 versus any other finding. Images were intensity-normalized, converted to grayscale and area-downsampled to 64×64. The benchmark has 924 images from ten sources (Table 1), and prevalence varies a lot between them (SD 0.274), so label shift matters.

Folds and freezing. Eight sources were each held out once; Figure1 and Hannover-DE, which have only 5 and 3 negative images, were used for training only. We assigned four to development (Actualmed, Italy, China, Spain) and four to confirmation (UK, Australia, Other, Unknown) before looking at any confirmation result; the split was not randomized. Every design decision used development folds only. At the end of iteration 8 we froze nine configurations and ran them once on the confirmation folds. Validation (a patient-level 15% split) and checkpoint choice (best source-validation AUROC) never touch the held-out site. One exception is descriptive: the label-free diagnostics were computed once over all eight sites in the first iteration; they involve no tuned model and no choice among methods used them.

Models and statistics. The encoder has four conv–BatchNorm–ReLU–max-pool blocks (widths 16–64), trained from scratch because no pretrained weights were reachable, with AdamW (learning rate 2×10^{-3}), batch size 64, 15 epochs, two seeds, on two CPU cores (25–80 s per run). The unit of generalization is the held-out site. For each method versus ERM we take the paired AUROC difference (seed-averaged) per site, then bootstrap hierarchically: resample sites, then resample test images within each chosen site stratified by class, using the same images for both methods (1500 replicates). This is wider than a bootstrap over sites alone, which matters because some sites are tiny (Australia has 6 COVID-19 images; China has 5 negatives). For label-shift and acquisition-shift experiments we also built a semi-synthetic benchmark: images are assigned at random to five virtual sites with known transforms (gamma, contrast, blur, noise, bias field) and controlled prior spread.

5 Results

Acquisition statistics leak the label. Five numbers with no anatomy predict the diagnosis on held-out sites with mean AUROC 0.662 (gradient boosting: 0.588; Fig. 1) and predict the source site with accuracy 0.367 against 0.224 for guessing the majority site. Re-measuring the leak on site-profile-augmented training images did not reduce it ($0.662 \rightarrow 0.660$), so the label-correlated signal seems to live mostly *within* sites (for example portable versus fixed acquisition, or patient severity) and not in site-level signatures that augmentation can randomize. Model scores track the same signal: within-class reliance is 0.438 for ERM and between 0.396 and 0.514 for the other methods. Site-profile augmentation and PAI lower it only slightly (0.396, 0.398), and the two methods with a class-conditional adversary and prior reweighting have the highest values (0.473, 0.514).

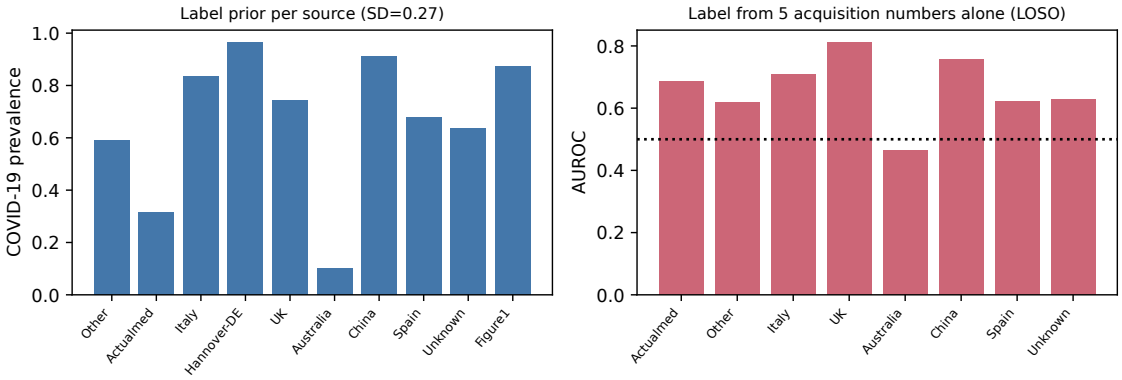


Figure 1: Left: COVID-19 prevalence per source. Right: AUROC for predicting the diagnosis on a held-out site from five acquisition statistics alone.

Method	AUROC dev	AUROC conf.	AUROC all	Worst fold	Δ vs ERM [95% CI]	Bal. acc.	ECE
ERM	0.758	0.707	0.733	0.565	–	0.634	0.157
GroupDRO	0.761	0.735	0.748	0.657	+0.015 [-0.015, +0.055]	0.689	0.144
DANN	0.752	0.702	0.727	0.645	-0.006 [-0.049, +0.035]	0.599	0.166
Generic aug.	0.748	0.716	0.732	0.619	-0.001 [-0.029, +0.032]	0.663	0.150
Site-profile aug.	0.757	0.694	0.726	0.641	-0.007 [-0.060, +0.038]	0.618	0.174
Profile adversary (PAI)	0.753	0.708	0.730	0.551	-0.002 [-0.020, +0.014]	0.615	0.154
SIDL-cc	0.722	0.699	0.711	0.635	-0.022 [-0.069, +0.030]	0.612	0.178
SIDL-phys	0.756	0.732	0.744	0.650	+0.011 [-0.016, +0.050]	0.656	0.192
SIDL-full	0.738	0.665	0.701	0.415	-0.031 [-0.085, +0.016]	0.590	0.175

Table 2: LOSO task metrics (mean over held-out sites of the two-seed mean). Δ is the paired AUROC difference to ERM over eight sites with hierarchical-bootstrap 95% interval. "Worst fold" is the lowest per-site AUROC. Balanced accuracy and ECE use the argmax decision.

No method significantly beats ERM. ERM reaches AUROC 0.758 on development folds and 0.707 on confirmation folds (Table 2, Fig. 2). Paired differences to ERM are small and all intervals include zero: GroupDRO +0.015 [−0.015, 0.055], SIDL-phys +0.011 [−0.016, 0.050], generic augmentation −0.001, PAI −0.002, DANN −0.006, site-profile augmentation −0.007, SIDL-cc −0.022, SIDL-full −0.031 [−0.085, 0.016]. Upper bounds rule out gains above about 0.055 for every method. Orderings are unstable: method means correlate only 0.500 (Spearman) between development and confirmation folds. Site-profile augmentation tied ERM on development folds (0.757 vs. 0.758) and fell below it on confirmation folds (0.694 vs. 0.707); SIDL-phys moved the other way. GroupDRO also has higher balanced accuracy (0.689 vs. 0.634) and a better worst-site AUROC (0.657 vs. 0.565), but ERM's worst site has only 6 positives, PAI does not share the gain, and balanced accuracy depends on an operating point that group reweighting shifts, so we treat both as exploratory. In an early iteration, whole-batch full-strength augmentation cost up to 0.083 AUROC on development folds (0.675 vs. 0.758); making it per-sample and mild removed the drop but never produced a gain.

Controlled label shift. On the semi-synthetic benchmark (identical image content, known transforms), DANN, cDANN and cDANN with prior reweighting all lose to ERM at the largest prior spread (0.4): −0.092 [−0.175, −0.016], −0.077 [−0.142, −0.019] and −0.047 [−0.092, −0.013]; reweighting alone is neutral (−0.000). The ordering matches theory (unconditional alignment hurts most, class-conditional with reweighting least), but at weight 0.5 the adversaries already lose 0.02–0.06 AUROC with no label shift at all. Lowering the weight moves them toward ERM and never above it, and the sign of the advantage of cDANN over DANN flips across weights; with six units per cell we cannot claim the class-conditional version helps (Table 3).

Conformal sets under-cover unseen sites. Sets calibrated on seen-site validation data miss the target (Table 4, Fig. 3): mean coverage is 0.836 (pooled) and 0.840 (site-balanced) at the 0.90 target, the worst site gets about 0.743, and 4 of 8 sites are more than five points short. The pattern shows up in both fold sets (0.829 on development, 0.843 on confirmation for pooled). Worst-site calibration restores the mean (0.904) but leaves two sites short (lowest 0.751, at "Other"). LSO-CV+ reaches mean coverage 0.942 and lowest site coverage 0.867 with no site more than five points short, at mean set size 1.747 against 1.389 for pooled. It over-covers on most sites; LSO-split lands nearer the target on average (0.905) but has a site at 0.732. Interpolated at 90% coverage, set size is between 1.574 and 1.603 for *every* calibration (pooled 1.582, LSO-CV+ 1.602). Development folds had suggested LSO-CV+ was about 4% smaller at matched coverage; the confirmation folds did not repeat that. So the calibrations differ in operating point, not efficiency, and LSO-CV+'s benefit is safety. Whether sets about 26%

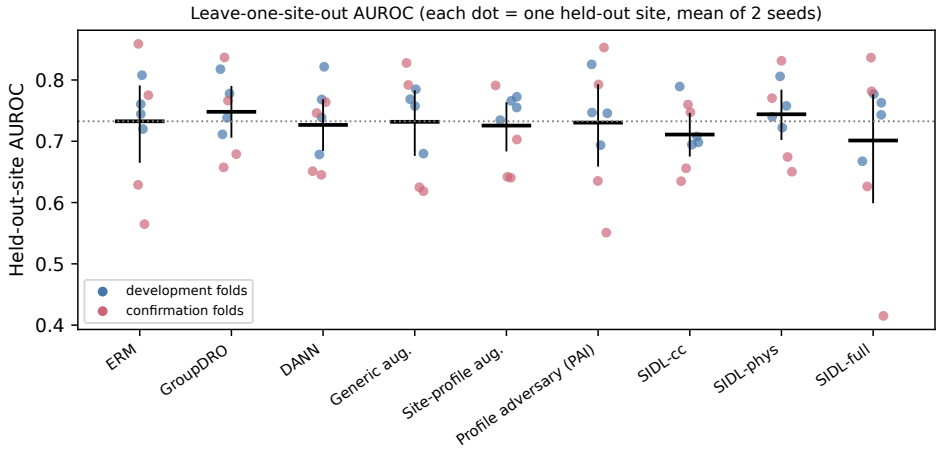


Figure 2: Held-out AUROC per method (dots: sites; bar and line: mean and bootstrap interval over sites).

Prior spread	ERM	ERM+prior	DANN	cDANN	cDANN+prior	Units
0.00	0.732	0.735	0.672	0.682	0.712	6
0.25	0.721	0.725	0.681	0.685	0.681	6
0.40	0.703	0.703	0.611	0.626	0.656	6

Table 3: Semi-synthetic label-shift sweep: held-out AUROC (virtual sites $V0, V2, V4$, two seeds, adversary weight 0.5).

larger are worth a six-point average shortfall (up to sixteen at the worst site) depends on the application. Two other ideas failed on development folds and were dropped: weighting calibration scores by a density ratio of the acquisition profile (coverage 0.834 vs. 0.829 for pooled), and correcting for estimated label shift [20, 24] (coverage 0.813 with an EM prior, 0.845 with the oracle prior; balanced accuracy got worse).

Why: validation optimism. For the ERM model we compared AUROC on seen-site validation data, the leave-site-out estimate (inner models on the sites they did not see), and the actual held-out AUROC (Fig. 4). Over 16 site-seed units, seen-site validation overstates held-out AUROC by 0.118 on average (absolute error 0.130); the leave-site-out estimate is nearly unbiased (0.007; absolute error 0.070). Neither ranks sites (Spearman with actual AUROC: -0.310 and -0.048): actual AUROC ranges from 0.56 to 0.86 while the leave-site-out estimate stays between 0.72 and 0.75, so it describes a typical unseen site, not a particular one. The per-site optimism gap and pooled-conformal coverage are rank-correlated in the expected direction (-0.452 over eight sites), consistent with optimism driving under-coverage but not significant at this sample size. Averaging the inner models (a free by-product) gives AUROC 0.748 vs. 0.733 for the single model (0.016 [-0.008, 0.049]), no significant gain.

Simulated shift is a poor stand-in. On the semi-synthetic benchmark with three acquisition-severity levels, seen-site pooled calibration already covers 0.896–0.922 at the 0.90 target, unlike the 0.836 on real sites. The LSO variants are no better, and LSO-split under-covers at the highest severity (0.834, lowest site 0.688). The real under-coverage therefore comes from differences between sites that our simulation does not contain (patient mix, case selection, labeling, image provenance). A calibration method checked only on simulated acquisition shift would have looked fine.

6 Discussion and Limitations

What we think the evidence supports. On this benchmark, augmentation, site adversaries, profile adversaries, CORAL and GroupDRO did not significantly improve held-out AUROC for a small CNN; gains above about 0.055 are unlikely. Acquisition statistics leak the label across sites and model scores follow them within class, but the methods that reduce this did not improve accuracy, so we cannot say the shortcut explains the accuracy loss. Seen-site validation is optimistic, standard conformal sets under-cover, and leave-site-out calibration avoids the worst shortfalls by paying in set size. We suspect invariance fails here because the data are tiny (sites of 52–207 images), the encoder is small and trained from scratch, the label-correlated signal is mostly within-site, and adversaries add optimization noise a small network cannot afford; we did not test these explanations.

Limitations. The study covers one modality and one task, although the title speaks of medical image diagnosis in general; our environment could not reach larger or credentialed datasets and had no GPU or pretrained weights, and we used no private data. The collection is retrospective and assembled from publications and case reports, so

Calibration	Cov.	Min cov.	Folds <0.85	Size	Cov. calib. err.	Frac. under	Size@90%
Pooled (seen-site val.)	0.836	0.743	4/8	1.39	0.048	0.389	1.58
Site-balanced (seen-site val.)	0.840	0.746	4/8	1.38	0.053	0.472	1.60
Worst-site (seen-site val.)	0.904	0.751	2/8	1.60	0.009	0.236	1.58
LSO-split	0.905	0.732	1/8	1.59	0.011	0.208	1.57
LSO-split, balanced	0.886	0.711	1/8	1.54	0.009	0.236	1.58
LSO-worst-site	0.960	0.832	1/8	1.81	0.083	0.028	1.59
LSO-CV+	0.942	0.867	0/8	1.75	0.041	0.069	1.60
LSO-CV+, balanced	0.926	0.792	1/8	1.69	0.020	0.153	1.60

Table 4: Conformal prediction for the ERM model on eight held-out sites, $\alpha = 0.1$ (target 0.90; two seeds). "Folds < 0.85" counts sites more than five points under target; the last three columns use the α grid. Rows 1–3 calibrate on seen-site validation data; rows 4–8 use leave-site-out scores.

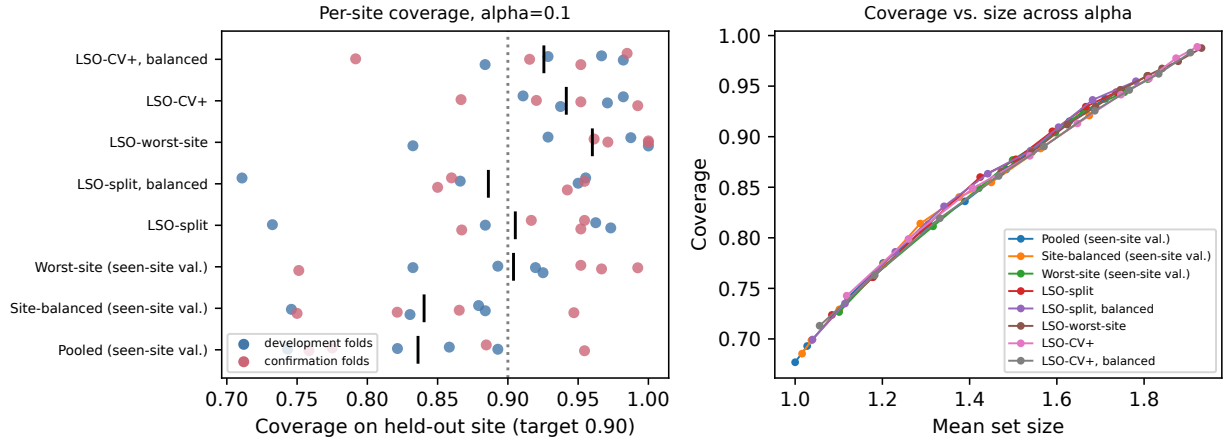


Figure 3: Left: coverage on each held-out site at $\alpha = 0.1$ (bar: mean). Right: coverage versus set size over the α grid; all calibrations lie on nearly one curve.

absolute AUROC is not clinically meaningful, and "Other" and "Unknown" mix institutions. With eight sites, two seeds, tiny test sets and no multiplicity correction, "no significant difference" is not evidence of equivalence below roughly 0.03 AUROC. The development/confirmation split was ours and not random, and tuning budgets were modest, so a better-tuned variant could still help. No leave-site-out conformal rule has a finite-sample guarantee under site shift, and coverage is marginal over images, not conditional on class or patient. The profile estimators are proxies. Larger data, stronger encoders and more sites would be needed to show that invariance helps.

Conclusion. An invariance recipe built from class-conditional alignment and site-profile augmentation did not beat ERM on held-out sites under a protocol that froze settings before the confirmation run. What lasted is a label-free leak diagnostic, a measured gap between seen-site validation and leave-site-out performance, the observation that standard conformal sets under-cover unseen sites and a site-fold CV+ rule avoids the worst cases at a cost in set size, and the finding that simulated acquisition shift does not reproduce the real effect.

Reproducibility and disclosure. Code, frozen configurations, per-run results and `scripts/final_report.py`, which regenerates every table, figure and headline number, are in the repository; `scripts/build_covid_cxr.py` rebuilds the benchmark from the three public collections, and `experiments/ITERATIONS.md` logs every experiment, including failures. Side analyses quoted in the text (the dropped conformal variants, the iteration-1 augmentation result, the leak-removal check) come from the scripts named in that log. Early iteration-1 runs on two confirmation sites were never used and are excluded. The code, experiments and a draft of this paper were produced with an AI assistant (Claude) under human direction and then edited for clarity and length; all numbers come from the released code and the authors are responsible for the content. Only public, de-identified images were used.

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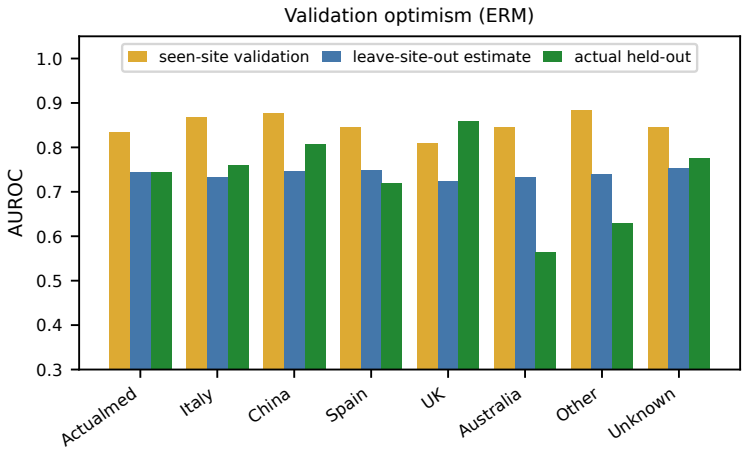


Figure 4: AUROC of the ERM model estimated on seen-site validation data, by leave-site-out models, and obtained on the held-out site (two-seed mean).

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Appendix: per-site AUROC

Method	Actualmed^d	Italy^d	China^d	Spain^d	UK^c	Australia^c	Other^c	Unknown^c
ERM	0.745	0.761	0.808	0.720	0.859	0.565	0.629	0.775
GroupDRO	0.778	0.738	0.818	0.711	0.837	0.679	0.657	0.766
DANN	0.768	0.739	0.822	0.678	0.746	0.651	0.645	0.764
Generic aug.	0.785	0.757	0.769	0.680	0.828	0.625	0.619	0.792
Site-profile aug.	0.755	0.766	0.773	0.735	0.703	0.642	0.641	0.791
Profile adversary (PAI)	0.747	0.746	0.825	0.694	0.853	0.551	0.635	0.793
SIDL-cc	0.698	0.789	0.708	0.694	0.747	0.656	0.635	0.760
SIDL-phys	0.740	0.758	0.806	0.722	0.831	0.674	0.650	0.770
SIDL-full	0.763	0.777	0.743	0.667	0.782	0.415	0.626	0.837

Table 5: Held-out AUROC per site (two-seed mean); d = development fold, c = confirmation fold.